1]Interactive Sales Dashboard:-

```
import streamlit as st
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [12000, 15000, 18000, 14000, 22000, 25000],
    "Profit": [2000, 2500, 3000, 1800, 4000, 5000],
    "Expenses": [8000, 9000, 10000, 8500, 12000, 13000],
    "Category": ["A", "B", "A", "C", "B", "C"]
}
df = pd.DataFrame(data)
st.title("Interactive Sales Dashboard")
st.sidebar.header("Filters")
selected_months = st.sidebar.multiselect(
    "Select Months",
    options=df["Month"].unique(),
    default=df["Month"].unique()
)
selected_category = st.sidebar.selectbox(
    "Select Category",
    options=["All"] + list(df["Category"].unique())
)
sales_threshold = st.sidebar.slider(
    "Minimum Sales",
    min_value=int(df["Sales"].min()),
    max_value=int(df["Sales"].max()),
    value=int(df["Sales"].min())
)
filtered_df = df[df["Month"].isin(selected_months)]
if selected_category != "All":
    filtered_df = filtered_df[filtered_df["Category"] == selected_category]
filtered_df = filtered_df[filtered_df["Sales"] >= sales_threshold]
st.dataframe(filtered_df)
st.subheader("Sales Trend")
fig, ax = plt.subplots()
ax.plot(filtered_df["Month"], filtered_df["Sales"], marker="o")
st.pyplot(fig)
st.subheader("Profit Chart")
fig, ax = plt.subplots()
ax.bar(filtered_df["Month"], filtered_df["Profit"])
st.pyplot(fig)
st.subheader("Sales vs Expenses")
fig, ax = plt.subplots()
ax.scatter(filtered_df["Sales"], filtered_df["Expenses"])
st.pyplot(fig)
```

Run with:
streamlit run Practice.py

2] Accessible Visualization with Sonification

```
import streamlit as st
import pandas as pd
import plotly.express as px
import pyttsx3
engine = pyttsx3.init()
data = {
    "Category": ["A", "B", "C", "D", "E"],
    "Value": [10, 25, 15, 30, 20]
}
df = pd.DataFrame(data)
st.title("Accessible Multimodal Visualization")
selected_category = st.selectbox("Select Category", df["Category"])
filtered_df = df[df["Category"] == selected_category]
fig = px.bar(
    filtered_df,
    x="Category",
    y="Value",
    color="Category",
    text="Value"
)
st.plotly_chart(fig)
value = int(filtered_df["Value"].values[0])
summary = f"Category {selected_category} has value {value}"
st.write(summary)
if st.button("Play Audio"):
    engine.say(summary)
    engine.runAndWait()
st.dataframe(df)
```

Run with:
streamlit run Sonification.py

3] Student Performance Dashboard

```
import streamlit as st
import pandas as pd
import matplotlib.pyplot as plt
st.title("Student Performance Dashboard")
data = {
    "Name": ["Amit", "Neha", "Rahul", "Sneha", "Kiran"],
    "Math": [85, 92, 70, 88, 60],
    "Science": [78, 88, 65, 90, 55],
    "English": [90, 95, 72, 85, 65]
}
df = pd.DataFrame(data)
st.dataframe(df)
subject = st.selectbox("Select Subject", ["Math", "Science", "English"])
min_marks = st.slider("Minimum Marks", 0, 100, 50)
filtered_df = df[df[subject] >= min_marks]
st.subheader("Histogram")
plt.hist(filtered_df[subject])
st.pyplot(plt)
st.subheader("Top Students")
plt.bar(filtered_df["Name"], filtered_df[subject])
st.pyplot(plt)
```

Run with:
streamlit run Practice1.py

4] Age Based Data Explorer

```
import streamlit as st
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
st.title("Age Based Data Explorer")
data = {
    "Age": [22, 25, 30, 35, 40, 28, 32, 45],
    "Salary": [25000, 30000, 40000, 50000, 60000, 35000, 45000, 65000],
    "Department": ["HR", "IT", "Finance", "IT", "HR", "Finance", "IT", "HR"]
}
df = pd.DataFrame(data)
age_range = st.slider("Age Range", int(df["Age"].min()), int(df["Age"].max()),
(20, 50))
dept = st.selectbox("Department", ["All"] + list(df["Department"].unique()))
filtered_df = df[(df["Age"] >= age_range[0]) & (df["Age"] <= age_range[1])]
if dept != "All":
    filtered_df = filtered_df[filtered_df["Department"] == dept]
plt.figure()
sns.boxplot(x="Department", y="Salary", data=filtered_df)
st.pyplot(plt)
```

Run with:
streamlit run Practice2.py

5] Dynamic Visualization Dashboard

```
import streamlit as st
import pandas as pd
import matplotlib.pyplot as plt
st.title("Dynamic Visualization Dashboard")
data = {
    "Name": ["Amit", "Neha", "Rahul", "Sneha", "Kiran"],
    "Math": [85, 92, 70, 88, 60],
    "Science": [78, 88, 65, 90, 55],
    "English": [90, 95, 72, 85, 65]
}
df = pd.DataFrame(data)
st.dataframe(df)
subject = st.selectbox("Subject", ["Math", "Science", "English"])
min_marks = st.slider("Minimum Marks", 0, 100, 50)
students = st.multiselect("Students", df["Name"], default=df["Name"])
filtered_df = df[(df["Name"].isin(students)) & (df[subject] >= min_marks)]
chart_type = st.selectbox("Chart Type", ["Bar", "Line"])
plt.figure()
if chart_type == "Bar":
    plt.bar(filtered_df["Name"], filtered_df[subject])
else:
    plt.plot(filtered_df["Name"], filtered_df[subject], marker='o')
st.pyplot(plt)
```

Run with:
streamlit run Practice3.py